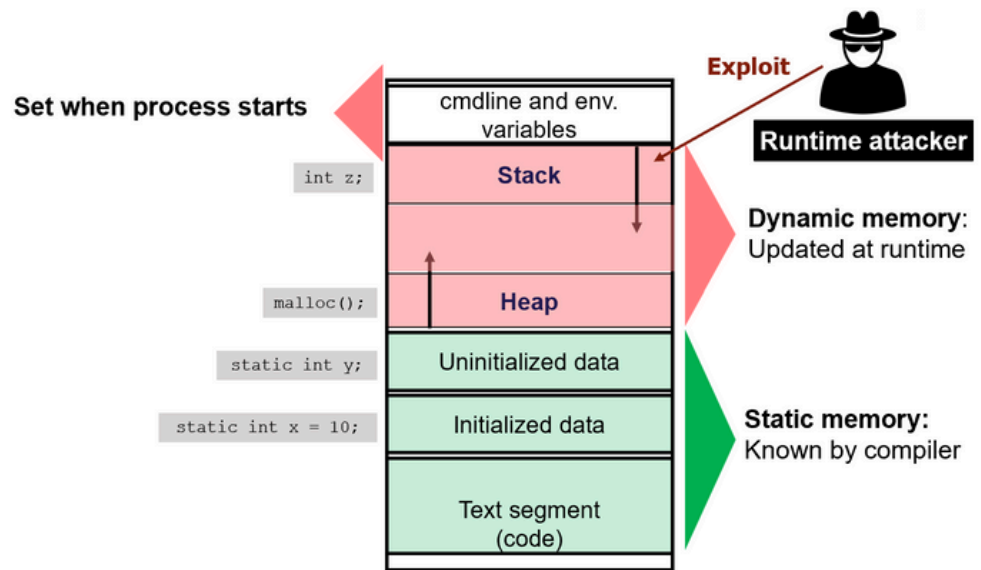


Runtime Software attacks



Description

Runtime attacks are sophisticated software attacks that try to compromise a device without injecting a new malicious code in the device. Instead, these attacks target the runtime memory of a device (e.g., RAM memory, as shown in the figure). Runtime attackers aim to manipulate the execution order of the code instructions at runtime, without adding a new code in the device. Therefore, such attacks remain often undetected.

This project will explore the underlying working mechanisms of runtime attacks. Students will build at least one proof-of-concept runtime attack in one device. Then, the project will propose a novel approach for detecting the runtime attacks. The main idea is to design an algorithm that traces the program execution flow at runtime. Additionally, there will be some evaluations about the effectiveness and efficiency of the proposed detection solution.

Goals

- Explore different techniques of runtime attackers
- Based on the identified mechanisms, construct a runtime attack in a single device
- Design a novel detection mechanisms for runtime attacks
- Evaluation of the performance and effectiveness